

TD - Practice - Solutions

Validating “identities”

For each of these “identities,” determine whether it is *valid* or *invalid* for all θ .

If valid, derive it from more basic sine-cosine identities.

Answers

1. $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$ — **Valid**. Fundamental double-angle form.
2. $\cos(2\theta) = 1 - 2\sin^2 \theta$ — **Valid**. Rearrangement of double-angle using $\cos^2 = 1 - \sin^2$.
3. $\cos(2\theta) = 2\cos^2 \theta + 1$ — **Invalid**. The correct form is $2\cos^2 \theta - 1$.
4. $\tan(2\theta) = \frac{2\tan \theta}{1 - \tan^2 \theta}$ — **Valid**. Standard tangent double-angle formula.
5. $\tan(2\theta) = \frac{2\tan \theta}{1 + \tan^2 \theta}$ — **Invalid**. Wrong sign in the denominator.
6. $\tan^2 \theta = \sec^2 \theta - 1$ — **Valid**. From $1 + \tan^2 \theta = \sec^2 \theta$.
7. $\csc^2 \theta = 1 - \cot^2 \theta$ — **Invalid**. Should be $1 + \cot^2 \theta = \csc^2 \theta$.
8. $\sin^2 \theta = 1 + \cos^2 \theta$ — **Invalid**. The correct form is $\sin^2 \theta + \cos^2 \theta = 1$.
9. $\sin(2\theta) = 2\sin \theta \cos \theta$ — **Valid**. Standard double-angle identity.
10. $\sin^2 \theta = \frac{1 + \cos(2\theta)}{2}$ — **Invalid**. Should be $\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}$.
11. $\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$ — **Valid**. Standard half-angle rewrite for $\cos^2 \theta$.
12. $\cos^2 \theta = \frac{1 - \cos(2\theta)}{2}$ — **Invalid**. Should be $\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$.
13. $\tan(\pi + \theta) = \tan \theta$ — **Valid**. Tangent has period π .
14. $\tan(\pi - \theta) = \tan \theta$ — **Invalid**. $\tan(\pi - \theta) = -\tan \theta$.
15. $\csc(-\theta) = -\csc \theta$ — **Valid**. Sine is odd so cosecant is odd.
16. $\sec(-\theta) = -\sec \theta$ — **Invalid**. Cosine is even so secant is even.
17. $\cot^2 \theta + 1 = \csc^2 \theta$ — **Valid**. Pythagorean identity for $\cot / \csc$.
18. $1 + \cot^2 \theta = \sec^2 \theta$ — **Invalid**. RHS should be $\csc^2 \theta$, not $\sec^2 \theta$.